THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

HP407 – Evidence Review and Synthesis for Decision Making

SUMMATIVE SUBMISSION TEMPLATE 2017/18

CANDIDATE NUMBER: *(**Five digit number** available via LSE for You – this is not the same as the number on your LSE ID card!)*	**8 2 5 7 4**
ASSESSMENT TITLE: *(E.g. Title of essay(s) attempted)*	**Systematic Review and Network Meta-analysis of Cognitive and/or Behavioral Therapies (CBT) in Tinnitus**
WORD COUNT: *Excludes references and appendices.*	**3278**

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ABBREVIATIONS

AAO – American Academy of Otolaryngology- Head and Neck Surgery

CBT – Cognitive Behavioral Therapy

CI – Confidence Interval

CPG – Clinical Practice Guidelines

HRQOL – Health related quality-of-life

iCBT – Internet-delivered cognitive-behavioral therapy

IF – Inconsistency Factor

ITT – Intention to treat analysis

N/A – Not available

NICE – National Institute for Health and Care Excellence

NMA – Network meta-analysis

RCT – Randomized Control Trial

SD – Standard Deviation

SMD – Standardised Mean Differences

TCT – Tinnitus Coping Training

VS - Versus

Systematic Review and Network Meta-analysis of Cognitive and/ or Behavioral Therapies (CBT) in Tinnitus

ABSTRACT

Objective: To evaluate the efficacy of cognitive and/or behavioral therapies in improving HRQOL, depression and anxiety associated with tinnitus.

Design: A pairwise meta-analysis of 12 RCTs with a total of 1,144 patients comparing cognitive and/or behavioral therapies to a wait-list control was performed for all outcomes (tinnitus HRQOL, depression and anxiety) with heterogeneity being explored with the use of subgroup analyses. NMA was performed on 19 studies with a total of 1,534 patients to compare cognitive and/or behavioral therapies to one another or to a waitlist control. Results were presented with SMDs and 95% CI.

Data Sources: EMBASE, MEDLINE, Pubmed, PsycINFO and the Cochrane Registry were used to identify studies.

Eligibility Criteria: RCTs comparing cognitive and/or behavioral therapies to one another or to a waitlist control were included. Participants over the age of 18 with tinnitus listed as the primary reason for treatment were included.

Results: Cognitive and/or behavioral therapies were associated with an improvement in tinnitus HRQOL (SMD 1.46; 95%CI 0.67, 2.24; I^2 95.3%), depression (SMD 0.95; 95%CI 0.2, 1.7; I^2 93.7%) and anxiety (SMD 1.85;95%CI -0.06, 3.75; I^2 97%), when compared to a waitlist control; however, the CI for the SMD related to anxiety indicates that results are not statistically significant. Though self-administered forms of CBT had larger effect sizes (SMD 3.44; 95%CI -.022,7.09; I^2 99%) on subgroup analysis of tinnitus HRQOL measures, only face-to-face CBT was shown to make a statistically significant improvement (SMD 0.75; 95%CI 0.53,0.97; I^2 0%). Overall, self-administered CBT had the highest likelihood of being ranked first in improving tinnitus HRQOL (75.1%), depression (82.7%) and anxiety (87.2%) but these results must be interpreted with caution given that the CIs negate statistical significance between most treatments.

Conclusions: Study results are consistent with previously published guidelines indicating that CBT is an effective form of therapy for tinnitus. The network meta-analysis provides new insights on different forms of cognitive-behavioral therapies for tinnitus. Future research should focus on performing larger studies with more participants, as well as assessing head-to-head comparisons of CBT delivery.

What is already known?

- Tinnitus is a chronic condition affecting up to 21% of the general population.
- Since no curative treatments exist, psychological therapies have been adopted to reduce the disabling secondary effects.
- CBT is an effective form of therapy for tinnitus in improving HRQOL, depression and anxiety.

What this study adds?

- Being the first network meta-analysis performed in tinnitus research, this study provides new important insights on the effects of different psychotherapeutic options.
- This study can be used to help inform recommendations on CBT included in the NICE tinnitus guidelines expected in 2020

INTRODUCTION

Tinnitus is a chronic condition defined as the perception of sound in the absence of an auditory stimuli (Jastreboff 2004). Tinnitus presents a major burden on health care systems with up to 21% of the general adult population being affected (Kim 2015). For many sufferers of chronic persistent tinnitus, disabling effects such as insomnia, social isolation, anxiety and depression can occur (Andersson 2009). Although a wide-range of therapies currently exist for the treatment of subjective tinnitus, none have proven to be curative.

Psychological therapies have been widely adopted due to evidence showing that disabling secondary effects depend more on psychological rather than acoustic properties (Cima 2014). One of the most widely applied psychological therapies to tinnitus management is cognitive behavioral therapy (CBT). CBT is an umbrella term which encompasses strategies from both the cognitive and behavioral therapeutic realms.

A *Cochrane Review* was performed in 2010 with the objective of assessing whether CBT was an effective therapy in the management of chronic tinnitus. Their meta-analysis identified eight RCTs with a total of 468 participants. They found that CBT, compared to no treatment, was not effective in reducing the primary outcome of subjective tinnitus loudness but did significantly improve the secondary outcomes including both quality of life and depression, suggesting that CBT had a positive impact on patient's ability to cope with tinnitus (Martinez-Devesa 2010). Limitations of this review were the broad definition used for CBT which included confounding interventions and the limited risk of bias assessment.

Subsequent AAO clinical practice guidelines (CPG) on tinnitus management adopted these suggestions and recommended CBT in patients with persistent tinnitus (Tunkel 2014). Though an updated *Cochrane Review* is planned to inform NICE on their tinnitus guidelines expected in 2020, the protocol design has received critique for potentially overemphasizing CBT for tinnitus with the current methodology (NICE 2017; Fuller 2017). Criticism surrounds the inclusion of various extended forms of psychotherapy into the definition of CBT, which has the potential to over-estimate the effects (Tyler 2017). Given that health systems have increasing pressures to provide cost-effective services, policy-makers depend on the results of rigorously designed systematic reviews to inform important decisions about service provision.

This review is important since it will attempt to stratify CBT by method of delivery and from other forms of closely related therapy, not planned by the *Cochrane Review* group. Furthermore, this review will attempt to update the literature by including RCTs that were not included in previous meta-analyses and will go one step further by attempting to complete the first network meta-analysis (NMA) of its kind in this therapeutic treatment area.

The objective of this study is to evaluate the efficacy of CBT compared to waitlist or other psychotherapies, for the treatment of tinnitus in adults.

METHODS

Study Protocol

A study protocol was developed and followed throughout the process.

Eligibility Criteria

Participants

Similar to the *Cochrane review* in 2011, adult patients, aged 18 years or older with tinnitus listed as the primary reason for treatment were included (Martinez-Devesa 2010).

Interventions

CBT was the primary intervention of interest. To avoid the potential for over-estimating treatment effects of CBT with too broad of a definition, the present review differed from the previous *Cochrane review* by stratifying studies into three groups: 'cognitive only', behavioral only' or 'mixed' cognitive-behavioral. Table 1 details the classification scheme employed.

Table 1. Classification of Interventions

Interventions	Definitions	Examples	References
'Cognitive only' Therapies	**Thought identification** Concerns the provision of education on negative automatic thoughts, teaching the patient how to identify their cognitive distortions. **Thought challenging** Concerns cognitive techniques that require change on the part of the patient, including the clinician instructing the patient on "thought stopping" exercises; "cognitive restructuring" exercises that is, teaching the patient to identify and modify or replace negative automatic thoughts; and having the patient role-play other perspectives using "Gestalt techniques".	1. Group Cognitive Therapy (GCT) 2. Cognitive restructuring (CR) 3. Attention control and imagery training (ACI)	Thompson 2017
'Behavioral only' Therapies	**Behavioral intervention** Concerns the use of behavioral techniques that require change on the part of the patient, including systematically increasing the patient's general level of activity using "behavioral activation" and "graded exposure" to tinnitus through silence or to noise as appropriate. **Relaxation** Concerns physical techniques designed to reduce autonomic arousal including progressive muscle relaxation and breathing exercises, typically entailing the tensing and relaxing of each muscle group in turn and diaphragmatic breathing or inhalation/exhalation-timed breathing, respectively.	1. Relaxation and distraction training (RD) 2. Relaxation and exposure training (RE) 3. Applied Relaxation Training (ART)	Thompson 2017 Goebel 2006
'Mixed' CBT	**Cognitive-behavioral therapy** Umbrella term for many different therapies that share some common elements from both behavioral and cognitive realms. **Acceptance and defusion** Concerns engaging in acceptance and cognitive defusion techniques, that is, to teach the patient to accept private experiences and to distance themselves from private events by attending more mindfully to the processes involved in thinking and feeling. **Mindfulness** Concerns the application of mindfulness meditation techniques.	1. CBT 2. Acceptance and commitment therapy (ACT) 3. Mindfulness 4. Cognitive behavioral tinnitus coping therapy. (TCT)	Thompson 2017

Comparators

Given that waitlist or no intervention was the most commonly identified control in a 2011 systematic review of CBT for tinnitus, it was used as the primary comparator for the pairwise meta-analysis (Fuller 2017). In the network meta-analysis (NMA), head to head studies comparing various forms psychotherapy to one another were also included (ie. 'mixed' CBT vs. 'cognitive only'). See appendix 1 for details on group stratification.

Outcomes and Prioritization

Primary Outcomes

The primary outcome in this review was improvement in validated tinnitus specific health-related quality of life (HRQOL) instruments. See appendix 2 for details.

If multiple measures of tinnitus specific HRQOL were used, a hierarchy of the outcome measures according to their psychometric validity was applied (Fackrell 2014). See Appendix 3 for details.

Secondary Outcomes

1. Reduction in depression.
2. Reduction in anxiety.

Secondary outcomes were measured using validated tools (appendix 4). All outcomes were measured when therapy was completed (usually six to eight weeks).

Studies

Similar to the previous *Cochrane review* on CBT for tinnitus, only RCTs were included in this review (Martinez-Devesa 2010; Ioannidis 2001). No restrictions were placed on year of publication or publication status.

Information Sources

We searched the following databases from their dates of inception to identify studies published in English, French or Spanish.

1. Cochrane Registry;
2. Ovid EMBASE;
3. Ovid MEDLINE;
4. Ovid PsycINFO;
5. PubMed.

Our search strategy was based on the protocol for the ongoing *Cochrane Review* of CBT for tinnitus (Fuller 2017). The adapted search strategy for each database can be found in Appendix 5.

Data Collection

Three authors independently screened titles and abstracts of all eligible studies and excluded irrelevant records. Full-text articles were obtained for studies that made it unclear from their abstracts if they were relevant. The three authors evenly divided the procurement of full-text articles and data extraction. At all times two of the listed authors were independently extracting data from the full-text articles into a standardized Excel (Microsoft, Redmond Washington) sheet developed by the authors *a priori*. Data items included information on: study source, participants characteristics, study methods, interventions, outcomes and results. See appendix 7 for a more in depth description. If articles were excluded at this stage because they did not meet eligibility criteria, reasons for exclusion were documented.

Risk of Bias

Any risk of bias with the potential of affecting cumulative evidence was considered.
Risk of bias assessment, as guided by the Cochrane risk-of-bias tool, was performed on all included studies by all three authors (Higgins 2011). Details can be found in appendix 8. To consider risk of bias across studies, particular attention was paid to publication bias. See appendix 9 for further detail.

Data Synthesis

Several analyses were performed on the extracted data. First, a qualitative analysis of the studies was undertaken, describing the study, the population and the methodological characteristics. See Table 2/3 for details.

Second, a pairwise meta-analysis was conducted using a random effects method, to compare CBT or other active forms of therapy, to a waitlist condition (Borenstein 2010). The data for primary and secondary outcomes was continuous. Treatment effects were measured using standardised mean differences (SMD) since studies assessed the same outcomes but measured them in a variety of ways, necessitating the conversion to a uniform scale before they could be combined. The results of the meta-analyses were presented in forest plots indicating SMDs with 95% confidence intervals (CI). I^2 was used to assess heterogeneity (Thorlund 2013). Meta-analyses were conducted on both primary and secondary outcomes (tinnitus HRQOL, depression and anxiety).

To further explore the heterogeneity (I^2) revealed in the meta-analyses, subgroup analyses were performed when heterogeneity was high. For all outcome measures, a subgroup analysis was performed by stratifying therapies into either 'mixed' CBT or other forms of therapy including cognitive or behavioral 'only'. The 'mixed' CBT group was further subdivided by method of delivery which included either face-to-face or self-administered therapies. For the primary outcome measure, a second subgroup analysis was performed on the self-administered 'mixed' CBT group based on intention to treat analysis (ITT).

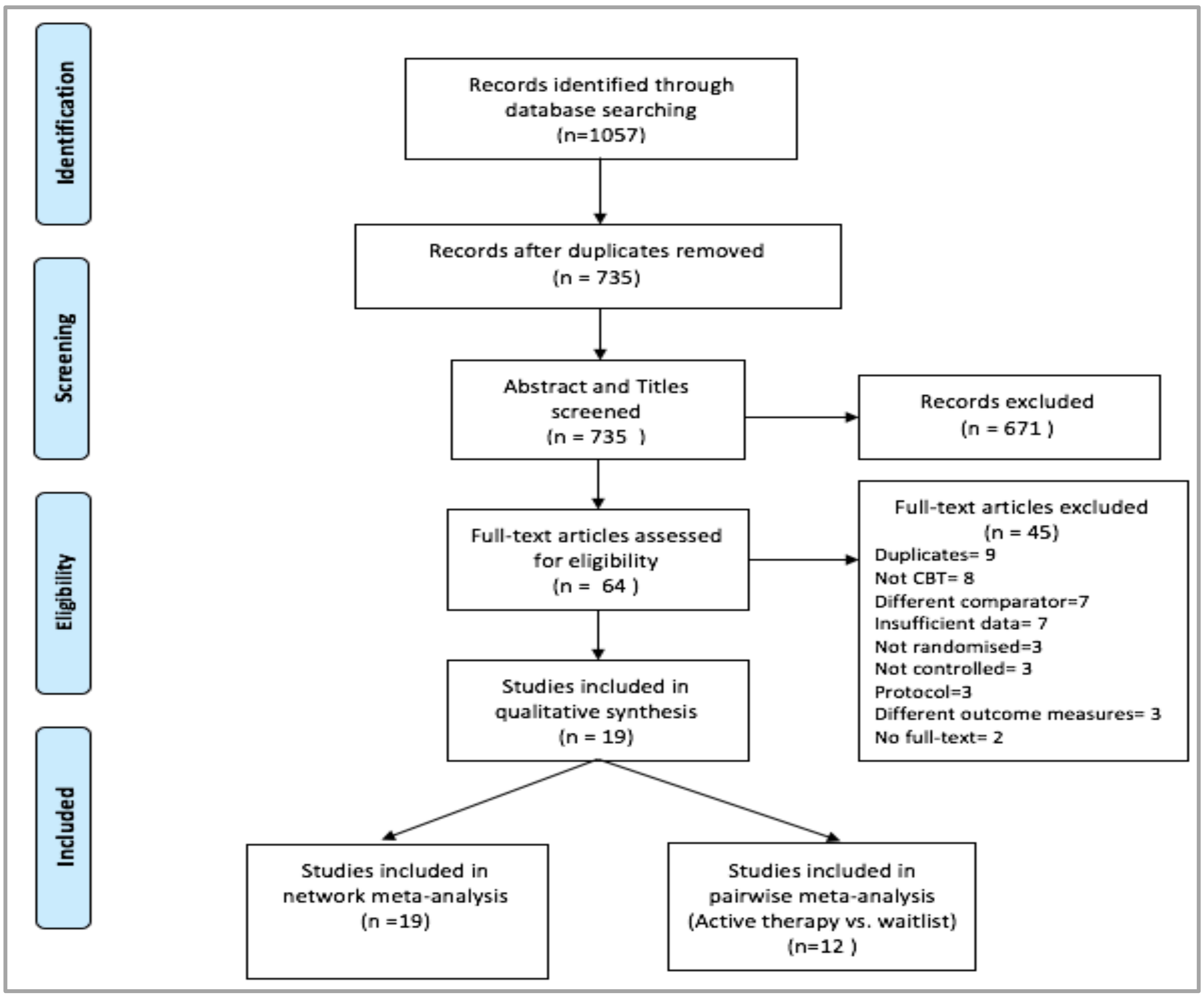
Finally, a third analysis was performed to conduct an NMA comparing the effects of all forms of cognitive and/or behavioral therapies in tinnitus, as this allowed us to compare the direct and indirect relationships between therapies (Caldwell 2005). Treatment effects for all three outcome measures were presented by network diagrams and interval plots indicating SMDs with 95% CI. From the NMA results, rankograms were also created for each outcome measure, demonstrating the distribution of ranking probabilities of the various therapies. Additionally, the inconsistency between direct and indirect results was assessed using inconsistency plots. The plots calculated both inconsistency factors (IF) and heterogeneity variance (τ^2) (Salanti 2014).

STATA (StatacCorp, Texas US) was used to conduct the meta-analysis, NMA, subgroup analyses and heterogeneity estimates.

RESULTS

Identified and eligible studies

Figure 1. PRISMA flow diagram of study selection process (Moher 2009)



Characteristics of included studies

We included 19 studies in the qualitative synthesis (Figure 1 and Table 2), totalling 1,534 patients. Figure 1 shows the study selection process and indicates reasons for exclusion. Characteristics of individual studies can be found in Tables 2 and 3.

Table 2. Selected characteristics of the studies[c]

Study	Sample size	Women (%)	Age (years)	Tinnitus Duration[a]	Attrition (%)[b]	Treatment Duration (weeks)	Type	Format	Control condition	Outcome Tinnitus	Depression	Anxiety
Andersson et al., 2005	23	48.0	70.1	13.0	0	6	CBT	1	0	TRQ	HADS	HADS
Andersson et al., 2002	117	46.0	47.9	6.3	18	6	iCBT	2	0	TRQ	HADS	HADS
Arif et al, 2017	61	59.0	56.1	5.9	41	15	Mindfulness Meditation Relaxation Therapy	1 4	1	TRQ	HADS	HADS
Davies et al, 1995	30	57.0	56.3	4.1	53	6	Cognitive Therapy Passive Relaxation Applied Relaxation	3 4 4	1	TEQ	BDI	n/a
Henry et al, 1998	54	38.0	56.3	n/a	20	8	Attention, Control and Imagery Therapy (ACI) Cognitive Restructuring Therapy (CT) ACI + CR	3 3 3	0	THQ	BDI	n/a
Hesser et al, 2012	99	43.4	48.5	0.8	10	8	iCBT iACT	2 2	1	THI	HADS	HADS
Ireland et al, 1985	33	53.3	55.9	5.3	9	1-4	Relaxation training with counterdemand instructions Relaxation training with neutral demand instructions	4 4	0	n/a	BDI	STAI
Jasper et al, 2014	128	39.8	51.2	8.5	6	10	Group CBT iCBT	1 2	1	THI	HADS	HADS
Kaldo et al, 2007	72	48.6	47.2	10.5	7	6	Bibliotherapy	2	0	THI	HADS	HADS

Kaldo et al, 2008	51	43.1	46.2	7.8	14	6	iCBT	2	1	THI	HADS	HADS
							Group CBT	1				
Kreuzer et al, 2012	36	47.2	50.7	n/a	14	9	Mindfulness and Body Psychotherapy	1	0	THI	BDI	n/a
Kroner et al, 2003	95	52.0	46.8	5.5	23	n/a	Tinnitus Coping Training	1	0	TQ	ADS	n/a
							Minimal Contact Intervention-Education	1				
							Minimal Contact Intervention Relaxation	4				
Malouff et al, 2009	162	44.4	57.6	n/a	23	8	Bibliotherapy	2	0	TRQ	n/a	n/a
McKenna et al, 2017	75	45.0	50.0	4.7	17	8	Mindfulness based cognitive therapy	1	1	TFI	HADS	HADS
							Intensive Relaxation	4				
Nyenhuis et al, 2013	304	45.4	48.5	0.3	32	n/a	Group CBT	1	0	TQ	PHQ-9	n/a
							iCBT	2				
							Bibliotherapy	2				
Philippot et al, 2012	30	40.0	60.0	n/a	17	7	Psycho Education + Mindfulness	1	1	QIPA	BDI	STAI
							Psychoeducation + Relaxation	4				
Weber et al, 2001	57	45.0	42.6	n/a	n/a	10	Progressive Muscle Relaxation	4	0	TQ	n/a	n/a
Westin et al, 2011	64	46.9	50.9	8.3	3	10	ACT	1	0	THI	HADS	HADS
Kroner et al, 1995	43	39.5	48.0	4.5	21	10	TCT Group 1 & Group2	1	0	TQ	VWB-D	n/a

Notes- Format: 1= Face-to-face 'mixed' CBT, 2=Self-administered 'mixed' CBT, 3=Cognitive 'only' therapy, 4=Behavioral 'only' therapy ; Control Condition: 0=Waiting list, 1=active control (other form of psychotherapy); TRQ= Tinnitus Reaction Questionnaire, TEQ= Tinnitus Effect Questionnaire, THQ= Tinnitus Handicap Questionnaire, THI= Tinnitus Handicap Inventory, TQ= Tinnitus Questionnaire, TFI= Tinnitus Functional Index; HADS= Hospital Anxiety and Depression Scale, BDI= Beck Depression Inventory, ADS= A Depression Scale, PHQ-9= Patient Health Questionnaire, VWB-D=Variables of well-being depression; STAI= State and Trait Anxiety Inventory; n/a= not available

a- Tinnitus duration (in years); b- attrition of patients randomized to psychological intervention.; c-See references below.

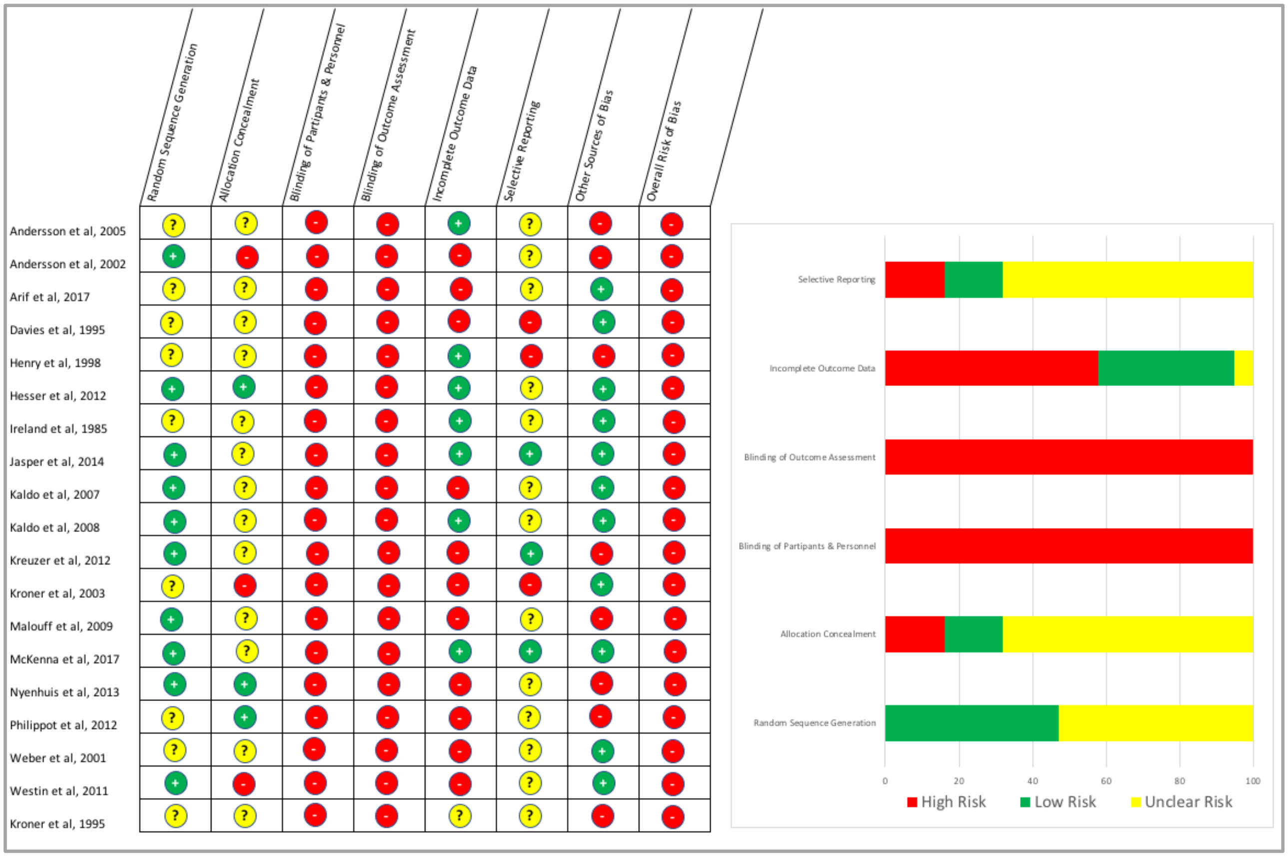
(Andersson 2002, 2005; Arif 2017; Davies 1995; Henry 1998; Hesser 2012; Ireland 1985; Jasper 2014; Kaldo 2007, 2008; Kreuzer 2012; Kröner-Herwig 2003; Malouff 2010; McKenna 2017; Philippot 2012; Nyenhuis 2013; Weber 2002; Kröner-Herwig 1995; Westin2011)

Table 3. Interventions included in the studies.

Study	Face to Face Mixed CBT	Self -administered Mixed CBT	Cognitive Only	Behavioral Only	Waitlist	Relevant Arms
Andersson et al, 2005	Group CBT				Waitlist	2
Andersson et al, 2002		iCBT			Waitlist	2
Arif et al, 2017	Mindfulness Meditation			Relaxation Therapy		2
Davies et al, 1995			Cognitive Therapy	Passive Relaxation Applied Relaxation		3
Henry et al, 1998			Attention, Control and Imagery Therapy (ACI)		Waitlist	4
			Cognitive Restructuring (CR)			
			ACI + CR			
Hesser et al, 2012		iCBT iACT				2
Ireland et al, 1985				Relaxation Training with Counter Demand Instructions	Waitlist	3
				Relaxation Training with Neutral Demand Instructions		
Jasper et al, 2014	Group CBT	iCBT				2
Kaldo et al, 2007		Bibliotherapy			Waitlist	2
Kaldo et al, 2008	Group CBT	iCBT				2

Kreuzer et al, 2012	Mindfulness and Body Psychotherapy			Waitlist	2
Kroner et al, 2003	Tinnitus Coping Training (TCT)		Minimal Contact Intervention Relaxation	Waitlist	3
Malouff et al, 2009		Bibliotherapy		Waitlist	2
McKenna et al, 2017	Mindfulness based cognitive therapy		Intensive Relaxation		2
Nyenhuis et al, 2013	Group CBT	iCBT		Waitlist	4
		Bibliotherapy			
Philippot et al, 2012	Psycho Education + Mindfulness		Psychoeducation + Relaxation		2
Weber et al, 2001			Progressive Muscle Relaxation	Waitlist	2
Westin et al, 2011	ACT			Waitlist	2
Kroner et al, 1995	TCT			Waitlist	2

Figure 2. Risk of bias within studies*



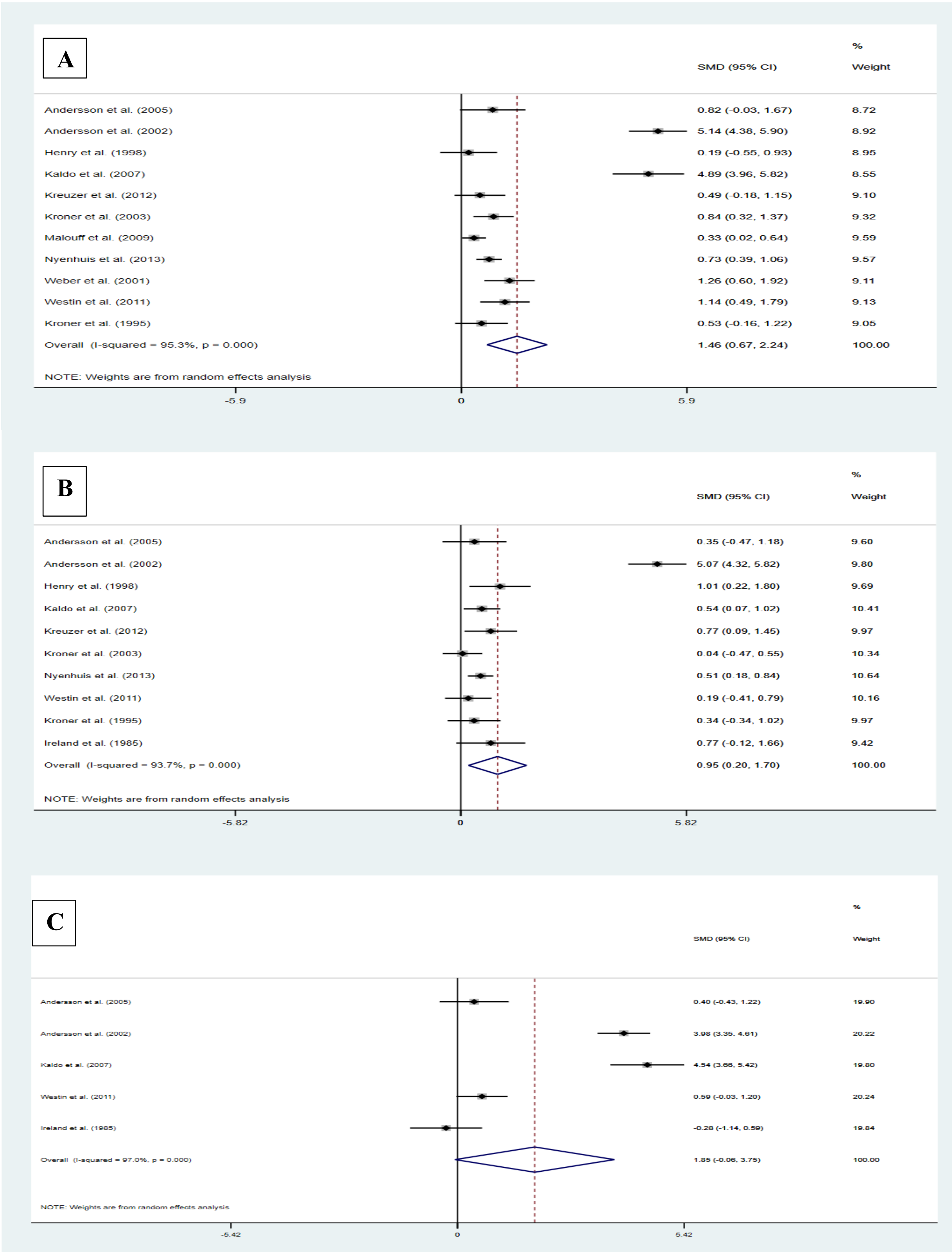
*Note – Bar graph horizontal axis is in percentages (%).

Pairwise meta-analyses results

12 studies were included in the pairwise meta-analysis which compared cognitive and/or behavioral therapies to a wait-list control, with a total of 1,144 patients.

For the primary outcome measure of tinnitus specific HRQOL, 11 studies were included in the meta-analysis with a total of 1,111 included patients. The secondary outcome measure of reduction in depression included 10 studies with a total of 925 patients. Lastly, the outcome measure evaluating reduction in anxiety included 5 studies, totalling 309 patients. Cognitive and/or behavioral therapies were associated with an improvement in the SMD of tinnitus specific HRQOL scores (SMD 1.46; 95% CI 0.67, 2.24), distress related to depression (SMD 0.95; 95% CI 0.2, 1.7) and distress related to anxiety (SMD 1.85;95% CI -0.06, 3.75), when compared to a waitlist control (Figure 3). Despite these favourable results, caution must be taken since the 95% CI for the SMD related to anxiety indicates that results are not statistically significant. Statistical heterogeneity was high in the analyses for tinnitus specific HRQOL (I^2 95.3%), depression (I^2 93.7%) and anxiety (I^2 97%).

Figure 3. Pairwise meta-analyses results on the effectiveness of cognitive and/or behavioral therapies compared to a wait-list control in improving: A) tinnitus specific HRQOL, B) depression, and C) anxiety.



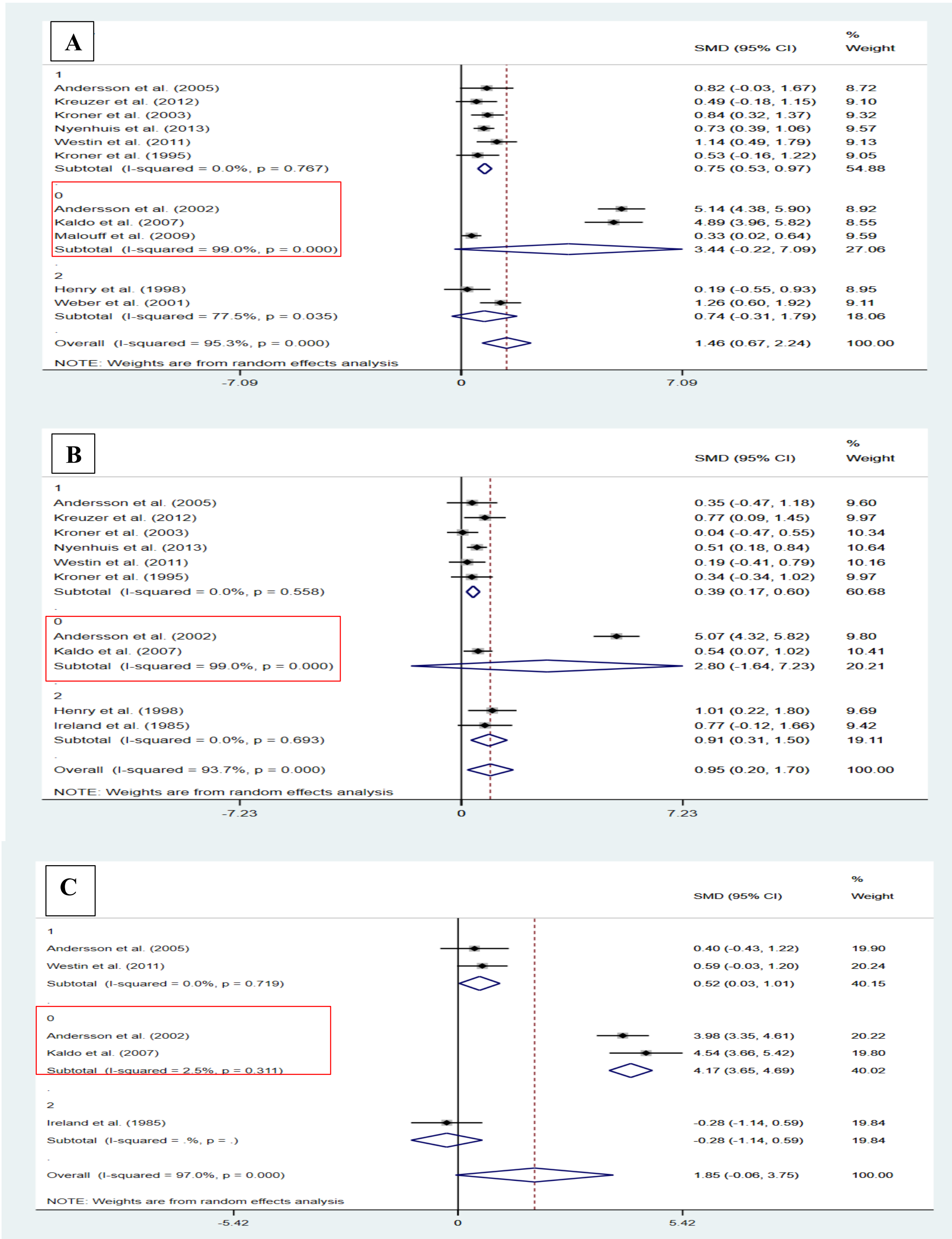
Subgroup analyses

At an attempt to explain the high level of heterogeneity found in the pairwise meta-analyses for all three outcome measures, subgroup analyses were performed by subdividing the 'mixed' CBT group, from cognitive or behavioral 'only' forms of therapy. In addition to this separation, the 'mixed' CBT group was further subdivided according to the route of administration of the therapy. Either face-to-face or self-administered delivery methods were possible.

The subgroup analyses for all three outcome measures generally helped to explain the heterogeneity found in the pairwise meta-analyses and suggested that the self-administered CBT group was likely behind the large heterogeneity observed (Figure 4).To illustrate this, the tinnitus specific HRQOL measure will be explored. In the pairwise meta-analysis for this outcome measure the overall heterogeneity was high (I^2 95.3%), subgroup analysis revealed low heterogeneity in the face-to-face 'mixed' CBT group with seven studies (I^2 0%), and high heterogeneity in both the self-administered 'mixed' CBT group with three studies (I^2 99%) and the cognitive or behavioral 'only' groups having only two studies (I^2 77.5%).

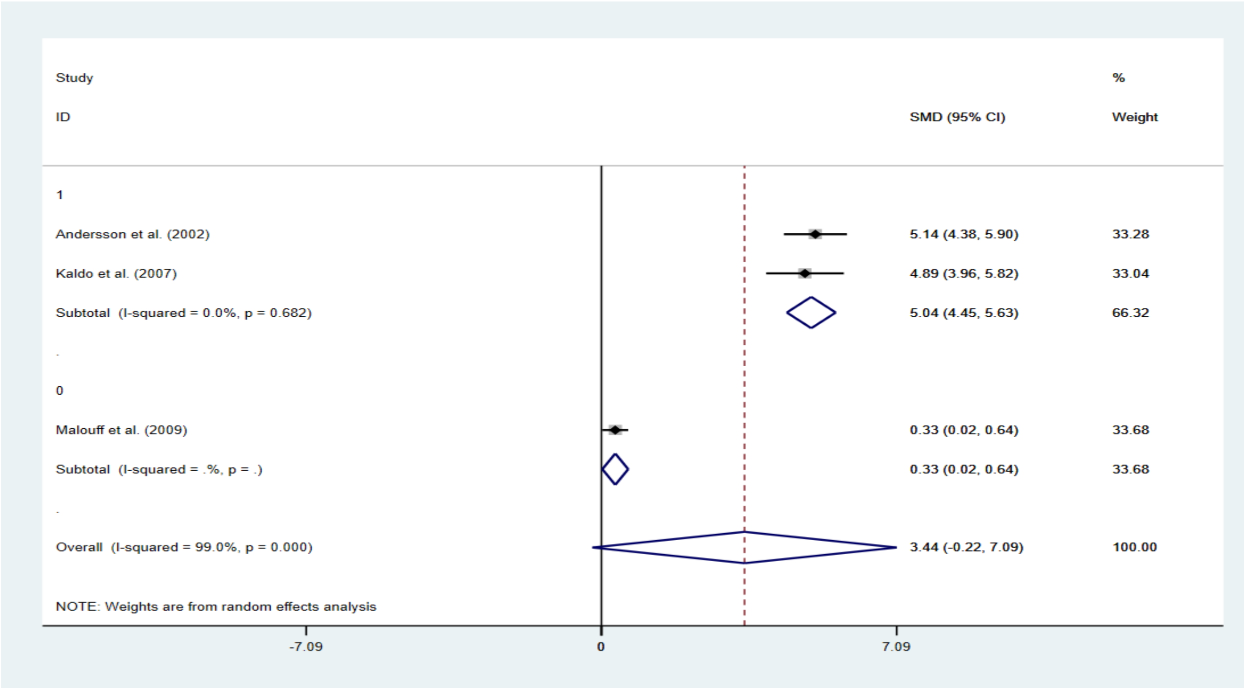
When observing Figure 4A, it is apparent that the results from Andersson 2002 and Kaldo 2007 are drastically different than other studies in the sample and could be driving the heterogeneity. To investigate this, a subgroup analysis was performed on the self-administered CBT group based on ITT analysis (Figure 5). This additional subgroup analysis is limited by having only 3 studies, but showed that in studies where ITT was performed that heterogeneity was low (I^2 0%) in comparison to studies not performing an ITT analysis. However interesting, the ITT analysis alone is unlikely to explain the large differences in effect size observed within the self-administered CBT group and other factors will be considered in the discussion.

Figure 4. Subgroup analyses on the results of the pairwise meta-analyses for A) tinnitus specific HRQOL, B) depression, and C) anxiety.



0- Self-administered CBT; 1-Group CBT; 2- Cognitive or Behavioral 'only' therapies

Figure 5. ITT subgroup analysis for self-administered CBT*



1- ITT performed; 0-Absence of ITT analysis

*Based on the results of the subgroup analysis for tinnitus specific HRQOL (see Figure 4 (A) 0)

Network meta-analyses results

A total of 19 studies were included in the NMA, with a total of 1,534 patients. In addition to the studies included in the meta-analysis which compared cognitive and/or behavioral therapies to waitlist control conditions, studies that compared cognitive and/or behavioral therapies to other forms of cognitive and/or behavioral therapies were also included (Figure 6).

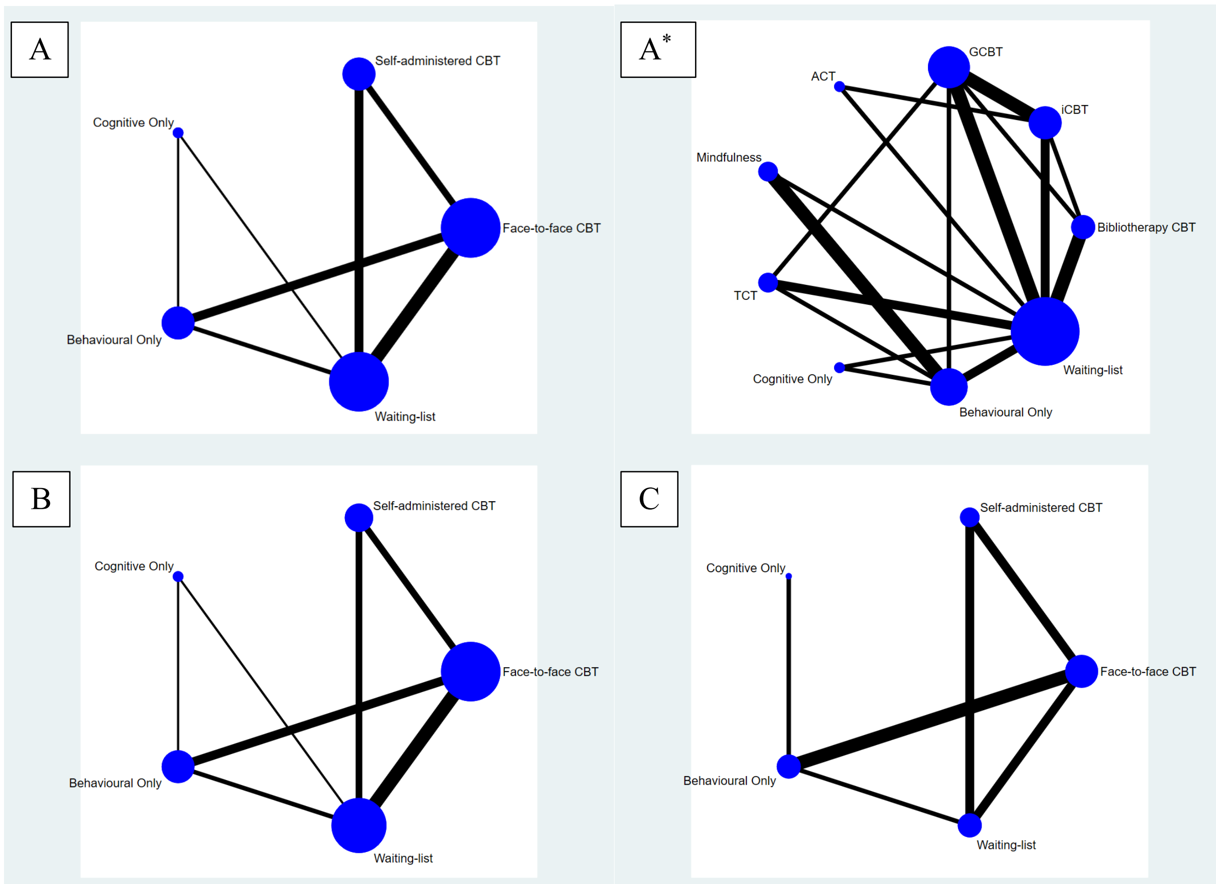
In the NMA for the improvement of tinnitus specific HRQOL, both self-administered and face-to-face CBT were shown to be superior to a waitlist control condition. Results for the remainder of the comparisons were not statistically significant (Figure 7A). An expanded network was formed using the previous outcome to stratify interventions by therapeutic similarities in treatment protocols (Figure 6A*). Statistically significant improvements were shown when bibliotherapy, internet-delivered CBT (iCBT) and group CBT were compared to a waitlist control (Figure 8), other comparisons were not statistically significant.

Moving on to the secondary outcome measures, depression scores were significantly improved when self-administered CBT was compared to both face-to-face CBT and waitlist controls; whilst anxiety scores were significantly improved when self-administered CBT was compared to face-to-face CBT, behavioral ‘only’ therapies, and waitlist controls. Face-to-face CBT was also shown to be superior when compared to a waitlist control. All other comparisons lacked statistical significance (Figure 7).

Overall, the NMA results suggest that self-administered CBT was the most effective treatment in improving tinnitus specific HRQOL (75.1%), depression (82.7%) and anxiety (87.2%) (Figure 9), these results must be interpreted with caution given that the CIs negate statistical significance between most treatments (Chaimani 2013).Within self-administered CBT, bibliotherapy (30%) had a slightly higher likelihood of being ranked first than iCBT (22%) in improving tinnitus HRQOL (Figure 10).

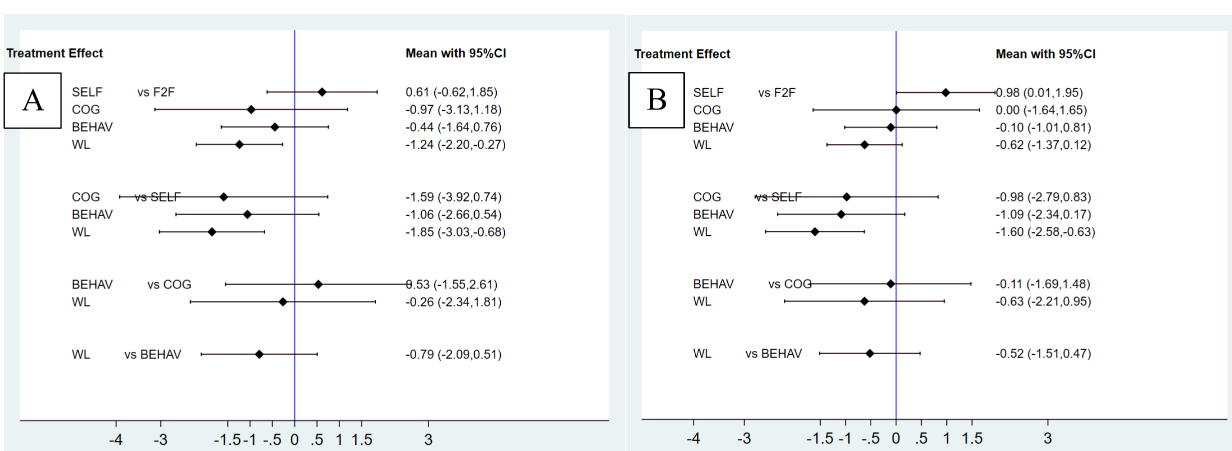
The inconsistency plots evaluated the statistical heterogeneity variance within the NMA and revealed high values across all three outcome measure when comparisons were performed for the F2F-SELF-WL loop (Figure 11-12), indicating that significant discrepancy between direct and indirect estimates exists (Salanti 2014).

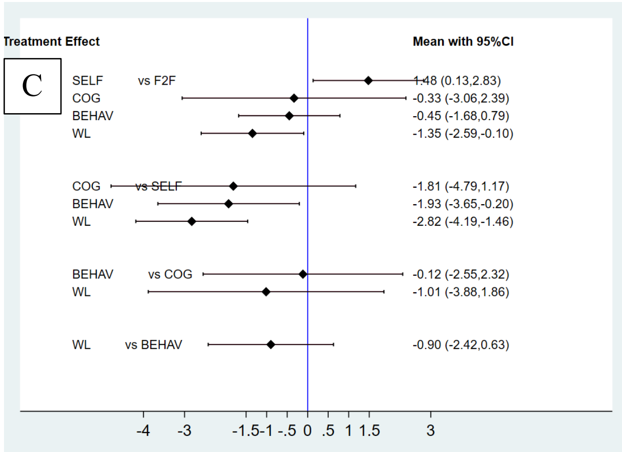
Figure 6. Network Diagrams showing the comparisons of various forms of cognitive and/or behavioral therapy in improving A) tinnitus specific HRQOL, B) depression and C) anxiety.



Note - *Represents an expanded network of A. Interventions have been further subdivided by therapeutic similarities in treatment protocols.

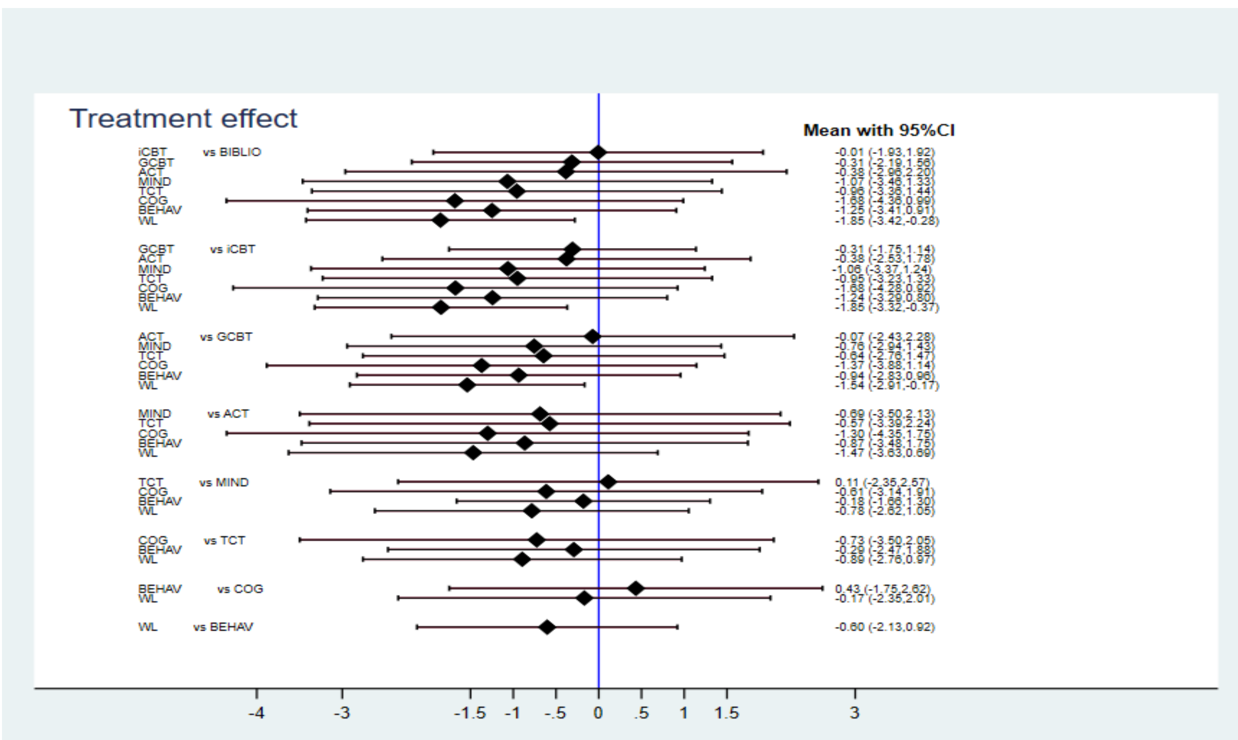
Figure 7. Interval plots of the NMA results on the effectiveness of cognitive and/or behavioral therapies in improving A) tinnitus specific HRQOL, B) depression, and C) anxiety.





Note – Treatment effect was measured by SMD, and not by means as indicated.

Figure 8. Interval plot of the network meta-analysis results on the effectiveness of cognitive and/or behavioral therapies in improving tinnitus specific HRQOL*.



Note - *Represents an expanded network of Figure 7 A. Interventions have been further subdivided by therapeutic similarities in treatment protocols. Treatment effect was measured by SMD, and not by means as indicated.

Figure 9. Rankings of the most effective cognitive and/or behavioral therapies in improving A) tinnitus specific HRQOL, B) depression, and C) anxiety.

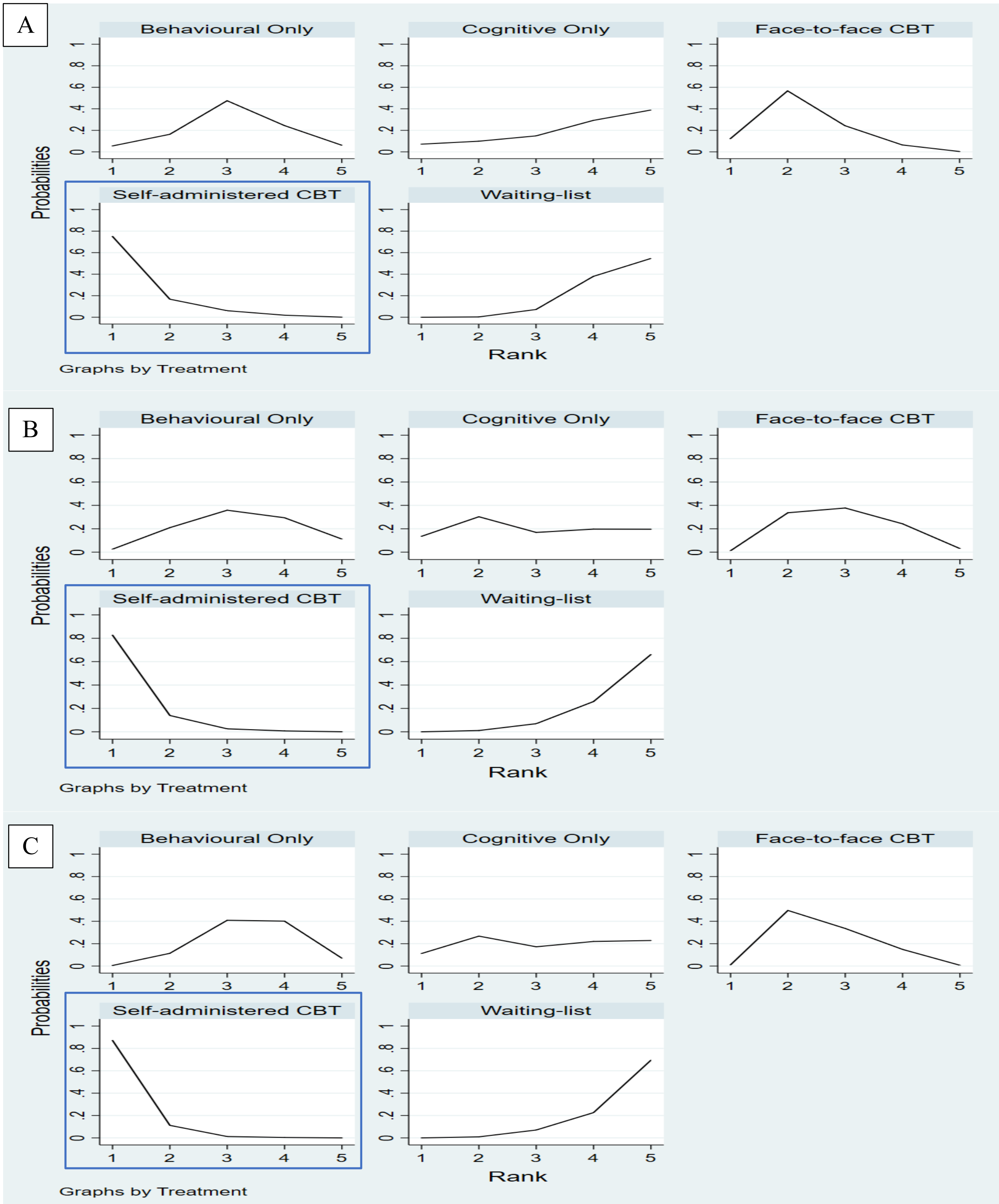
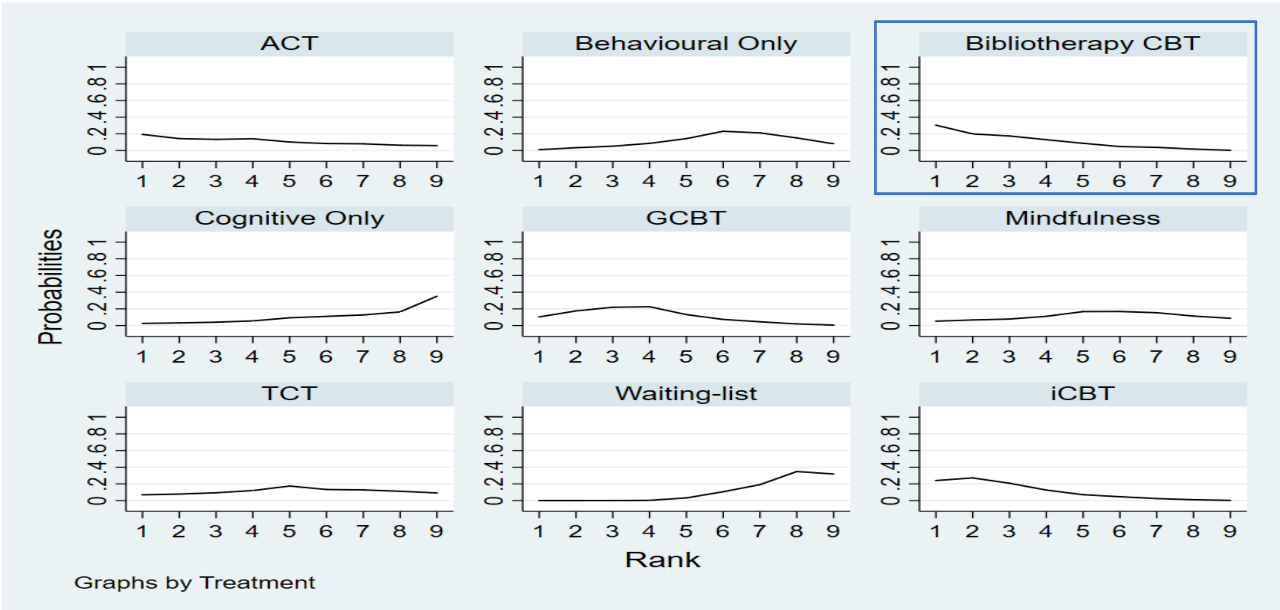


Figure 10. Rankings of the most effective cognitive and/or behavioral therapies in improving tinnitus specific HRQOL*.



Note - *Represents an expanded rankogram of Figure 9 A. Interventions have been further subdivided by therapeutic similarities in treatment protocols.

Figure 11. Inconsistency plots that show the heterogeneity between direct and indirect evidence in the network A) tinnitus specific HRQOL, B) depression, and C) anxiety.

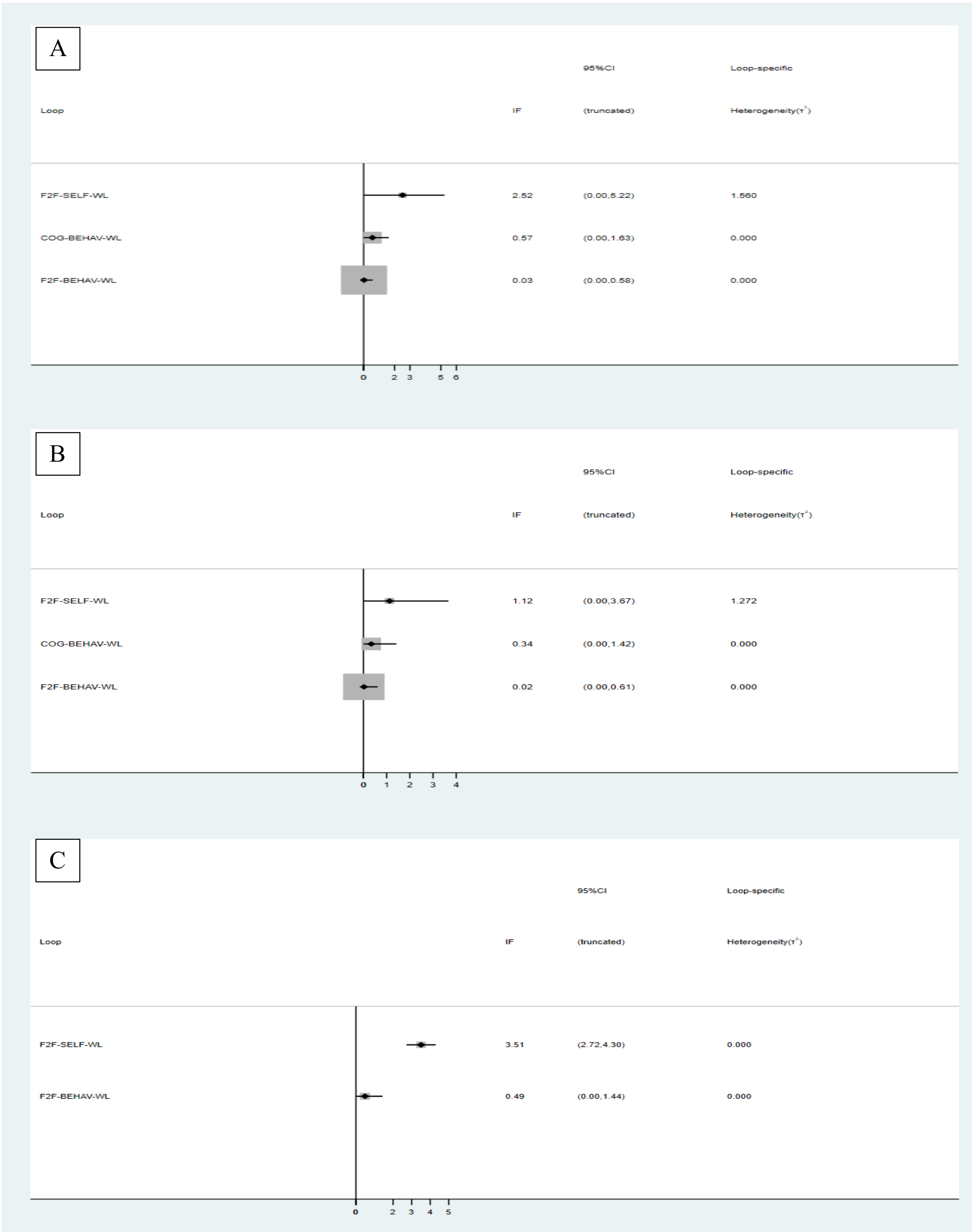
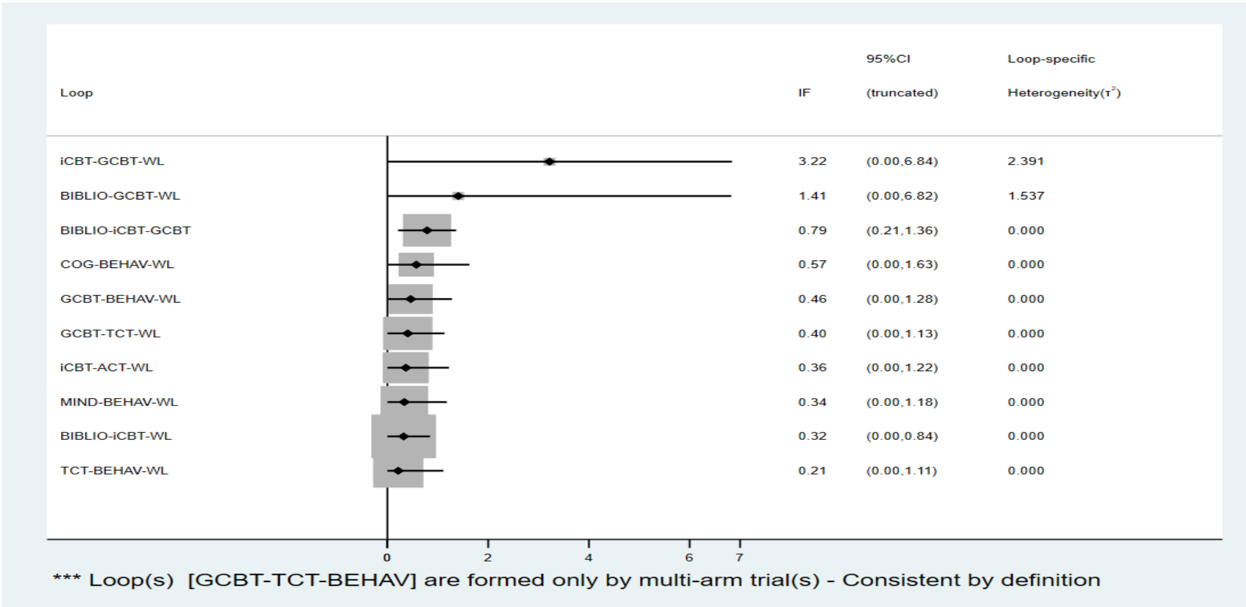


Figure 12. Inconsistency plots that show the heterogeneity between direct and indirect evidence in the network for tinnitus specific HRQOL*.



Note - *Represents an expanded inconsistency plot of Figure 11 A. Interventions have been further subdivided by therapeutic similarities in treatment protocols.

DISCUSSION

Cognitive and/or behavioural therapies were associated with an improvement in tinnitus HRQOL (SMD 1.46; 95%CI 0.67, 2.24), depression (SMD 0.95; 95%CI 0.2, 1.7) and anxiety (SMD 1.85;95%CI -0.06, 3.75) when compared to a waitlist control in pairwise meta-analyses. The results of the NMA from 19 RCTS on 1,534 patients, suggests that self-administered CBT had the highest likelihood of being ranked first in improving tinnitus specific HRQOL (78%), depression (82%) and anxiety (84%) caused by tinnitus distress. All results must be interpreted cautiously given that the CIs indicate that no statistical significance exists between most treatments.

Existing guidelines from the American Academy of Otolaryngology (AAO) are consistent with the findings of this study and suggest that CBT should be offered to patients with persistent tinnitus (Tunkel 2014). Though the 2007 Cochrane Review did not find that CBT had a significant impact on reducing tinnitus loudness, they also recommended CBT because of the improvements observed in tinnitus HRQOL, depression and anxiety (Martinez Devesa 2007). Despite favourable recommendations for the use of CBT from several organizations, many critiqued the findings of the 2007 *Cochrane Review*, as they questioned if the broad definition used for CBT had potentially over-estimated the therapeutic effects (Tyler 2017). This study aimed to rectify these concerns by stratifying treatments into therapeutically similar categories (Table 1) in the NMA, and has now appeased the critics by reconfirming *Cochrane's* findings. Given that studies from a variety of different countries (Sweden, the UK, Australia, Germany and Belgium) were included, it improves the generalisability of results.

This study provides new insights into the methods of delivery of CBT, and suggests that self-administered therapies may be superior to face-to-face formats. These results are informative, not indicative however, and must be interpreted carefully given that only face-to-face CBT was shown to make a statistically significant improvement in tinnitus HRQOL (SMD 0.75; 95%CI 0.53,0.97). Reasons for the larger effect sizes observed with self-administered CBT (SMD 3.44; 95%CI -.022,7.09) may be due to fundamental differences in the patient populations. It has been well documented, that patients who volunteer to take part in self-administered trials may be more inherently motivated which can create biased results and is one of the potential reasons for the high heterogeneity observed in the inconsistency plots for the F2F-SELF-WL loop (Bowling

2005). Nevertheless, because self-administered methods of delivery are much less resource intensive than face-to-face methods and could have significant policy implications for service delivery, they should be explored further.

Although our study was conducted thoroughly, several limitations that could affect our results exist. As has been previously documented by Hesser, there was a trend that type of control condition was associated with effect size, with studies using active controls having lower effect-sizes than studies using waitlist controls (Hesser 2011; Hesser 2011). Psychotherapies generally have larger effect sizes when compared to a waitlist control group, than when compared to other psychotherapeutic controls (Grissom 1996). For this reason, it would have been better to create an NMA with all currently available therapies for tinnitus, so that more accurate and comprehensive recommendations could be made on the best available treatments (Savage 2014). Furthermore, long-term effects of therapy have not been explored in this study, making it difficult to draw conclusions on sustained effects (Hesser 2011). Finally, the high risk of bias attributed to all studies is perhaps unfair, as it’s virtually impossible to remove the biases of psychotherapeutic RCTs that arise from the lack of effective blinding (Button 2015).

Larger studies with more participants are needed to establish more robust conclusions. Given the suggested superiority of self-administered CBT over face-to-face delivery, head-to-head studies should be designed to further assess this relationship, as this review only included three studies with such comparisons. The future of tinnitus research is likely to change dramatically over the next decade due to promising advances in genotyping. Tinnitus biomarkers are actively being developed and could help revolutionize the field by providing objective outcome measures to evaluate efficacy in future clinical trials (Haider 2017).

CONCLUSION

Our study results are consistent with previous meta-analyses and the AAO CPG indicating that CBT is effective form of therapy for tinnitus (Tunkel 2014; Martinez-Devesa 2010; Hesser 2011). This network meta-analysis is the first of its kind in this therapeutic area and provides new insights on the effects of different forms of cognitive and/or behavioral therapies for tinnitus. Future research should focus on performing larger studies with more participants, as well as assessing head-to-head comparisons of CBT delivery to strengthen the NMA.

PRIMA CHECKLIST – See appendix 9.

INDIVIDUAL CONTRIBUTION

The workload was distributed evenly amongst group members and the group functioned well in a very collegial and professional manor. The author of this paper was involved in *study design* (establishing inclusion/ exclusion criteria, determining primary/ secondary outcome measures and tools for prioritization, classifying interventions, finalizing search strategies and creating the data extraction form), *data extraction* (procured 1/3 of required articles, reviewed 2/3 of all titles/abstracts and full text articles, extracted data and performed risk of bias assessment on 2/3 of the articles, made PRISMA flow chart), *data analysis* (helped prepare the data for meta-analysis and NMA in excel by standardizing mean differences and calculating SDs, helped with analysis using STATA and formatting necessary figures), the individual *writing of this manuscript* and making the final *poster presentation*.

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APPENDICES

Appendix 1. Stratification of Studies

Studies will be stratified into one of six groups for comparison:

1. 'Mixed' CBT versus waitlist or no intervention as a control.
2. 'Cognitive only' therapies versus waitlist or no intervention as a control;
3. 'Behavioral only' therapies versus waitlist or no intervention as a control;
4. 'Mixed' CBT versus 'Cognitive only' therapy as a control;
5. 'Mixed' CBT versus 'Behavioral only' therapy as a control;
6. 'Cognitive only' therapy versus 'Behavioral only' therapy as a control;

Appendix 2. Tinnitus Outcome Measurement Tools and Citations

1. Tinnitus Questionnaire (Hallam 1988, Hallam 2008).
2. Tinnitus Functional Index (Meikle 2012).
3. Tinnitus Handicap Inventory (Newman 1996).
4. Tinnitus Handicap Questionnaire (Kuk 1990).
5. Tinnitus Reaction Questionnaire (Wilson 1991).
6. Tinnitus Severity Scale (Sweetow 1990).
7. Tinnitus Disability Index (Cima 2011).

For illustrative purposes an example of the Tinnitus Function Index (Meikle 2012) adapted by the Oregon Health & Sciences University in 2008 is provided below.

TINNITUS FUNCTIONAL INDEX

Today's Date ____________________ (Month / Day / Year) Your Name ______________________________ (Please Print)

Please read each question below carefully. To answer a question, select *ONE* of the numbers that is listed for that question, and draw a *CIRCLE* around it like this: 10% or 1.

I — Over the PAST WEEK...

1. What percentage of your time awake were you consciously **AWARE OF** your tinnitus?
   Never aware ► 0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100% ◄ *Always aware*

2. How **STRONG** or **LOUD** was your tinnitus?
   Not at all strong or loud ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Extremely strong or loud*

3. What percentage of your time awake were you **ANNOYED** by your tinnitus?
   None of the time ► 0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100% ◄ *All of the time*

SC — Over the PAST WEEK...

4. Did you feel **IN CONTROL** in regard to your tinnitus?
   Very much in control ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Never in control*

5. How easy was it for you to **COPE** with your tinnitus?
   Very easy to cope ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Impossible to cope*

6. How easy was it for you to **IGNORE** your tinnitus?
   Very easy to ignore ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Impossible to ignore*

C — Over the PAST WEEK...

7. Your ability to **CONCENTRATE**?
   Did not interfere ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Completely interfered*

8. Your ability to **THINK CLEARLY**?
   Did not interfere ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Completely interfered*

9. Your ability to **FOCUS ATTENTION** on other things besides your tinnitus?
   Did not interfere ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Completely interfered*

SL — Over the PAST WEEK...

10. How often did your tinnitus make it difficult to **FALL ASLEEP** or **STAY ASLEEP**?
    Never had difficulty ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Always had difficulty*

11. How often did your tinnitus cause you difficulty in getting **AS MUCH SLEEP** as you needed?
    Never had difficulty ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Always had difficulty*

12. How much of the time did your tinnitus keep you from **SLEEPING** as **DEEPLY** or as **PEACEFULLY** as you would have liked?
    None of the time ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *All of the time*

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TINNITUS FUNCTIONAL INDEX PAGE 2

Please read each question below carefully. To answer a question, select *ONE* of the numbers that is listed for that question, and draw a *CIRCLE* around it like this: 10% or 1.

A	**Over the PAST WEEK, how much has your tinnitus interfered with...**	*Did not interfere*										*Completely interfered*
	13. Your ability to **HEAR CLEARLY**?	0	1	2	3	4	5	6	7	8	9	10
	14. Your ability to **UNDERSTAND PEOPLE** who are talking?	0	1	2	3	4	5	6	7	8	9	10
	15. Your ability to **FOLLOW CONVERSATIONS** in a group or at meetings?	0	1	2	3	4	5	6	7	8	9	10
R	**Over the PAST WEEK, how much has your tinnitus interfered with...**	*Did not interfere*										*Completely interfered*
	16. Your **QUIET RESTING ACTIVITIES**?	0	1	2	3	4	5	6	7	8	9	10
	17. Your ability to **RELAX**?	0	1	2	3	4	5	6	7	8	9	10
	18. Your ability to enjoy **"PEACE AND QUIET"**?	0	1	2	3	4	5	6	7	8	9	10
Q	**Over the PAST WEEK, how much has your tinnitus interfered with...**	*Did not interfere*										*Completely interfered*
	19. Your enjoyment of **SOCIAL ACTIVITIES**?	0	1	2	3	4	5	6	7	8	9	10
	20. Your **ENJOYMENT OF LIFE**?	0	1	2	3	4	5	6	7	8	9	10
	21. Your **RELATIONSHIPS** with family, friends and other people?	0	1	2	3	4	5	6	7	8	9	10

22. How often did your tinnitus cause you to have difficulty performing your **WORK OR OTHER TASKS**, such as home maintenance, school work, or caring for children or others?

Never had difficulty ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Always had difficulty*

E Over the PAST WEEK...

23. How **ANXIOUS** or **WORRIED** has your tinnitus made you feel?

Not at all anxious or worried ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Extremely anxious or worried*

24. How **BOTHERED** or **UPSET** have you been because of your tinnitus?

Not at all bothered or upset ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Extremely bothered or upset*

25. How **DEPRESSED** were you because of your tinnitus?

Not at all depressed ► 0 1 2 3 4 5 6 7 8 9 10 ◄ *Extremely depressed*

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TFI scores range between 0-100 with higher scores indicating more severe impairment due to tinnitus. TFI scores can be stratified into three levels and interpreted as the following:

- <25 = mild tinnitus (little or no need for intervention);
- 25-50= significant problems with tinnitus (possible need for intervention); and
- >50 = tinnitus severe enough to qualify for more aggressive intervention.

Appendix 3. Tinnitus Outcome Measure Hierarchy

Fackrell et al. carried out an in-depth assessment of the development and validation of the multiple tools used to measure tinnitus outcomes (Fackrell 2014).

The proposed hierarchy according to psychometric validity:

1. Tinnitus Functional Index
2. Tinnitus Handicap Inventory
3. Tinnitus Handicap Questionnaire
4. Tinnitus Questionnaire
5. Tinnitus Reaction Questionnaire
6. Tinnitus Disability Index
7. Tinnitus Severity Scale

Appendix 4. Validated tools used in Secondary Outcomes

Secondary Outcomes

1. Reduction in depression as measured by validated depression questionnaires. Such questionnaires include the Beck Depression Inventory II (Beck 1996) and the Hamilton Rating Scale for Depression (Hamilton 1960).
2. Reduction in anxiety as measured by a validated anxiety scale. Examples include the Anxiety Sensitivity Index (Reiss1986) and the Beck Anxiety Inventory (Beck 1988).

Appendix 5. Search Strategies

Search strategies for databases were modelled on previously published search strategies from the Cochrane review on CBT for Tinnitus from 2010 (Martinez-Devesa et al. 2010) and adapted to incorporate the newly proposed search strategy published in the 2017 Cochrane protocol on CBT for tinnitus (Tunkel et al. 2014; Martinez-Devesa et al. 2010; Hesser, Weise, Westin, et al. 2011). Search strategies used to identify randomized control trials were taken from the *Cochrane Handbook for Systematic Reviews of Interventions* for each of the databases used (Higgins 2011).

Cochrane Search Strategy

#1 MeSH descriptor: [Tinnitus] explode all trees
#2 (tinnit*):ti,ab,kw
#3 #1 or #2
#4 MeSH descriptor: [Behavior Therapy] explode all trees
#5 MeSH descriptor: [Adaptation, Psychological] explode all trees
#6 MeSH descriptor: [Meditation] explode all trees
#7 (CBT or ACT or mindfulness orMBTR orMBSR orMBTSR or psychoeducation or iACT or iCBT or GCBT):ti,ab,kw
#8 ((cogniti* or relaxation or acceptance or commitment or adaptation) near (therap* or behavior* or behaviour* or strateg* or intervention* or approach* or psychotherap* or training or treatment or technique* or program* or counseling or counselling)):ti,ab,kw
#9 ((behaviour* or behavior* ormeditation) near (strateg* or intervention* or therap* or approach* or psychotherap* or technique* or counseling or counselling)):ti,ab,kw
#10 desensiti* and psychology*
#11 interpersonal therapy
#12 #4 or #5 or #6 or #7 or #8 or #9 or #10 or #11
#13 #3 and #12
#14 randomized controlled trial [pt]
#15 controlled clinical trial [pt]
#16 randomized [tiab]
#17 placebo [tiab]
#18 drug therapy [sh]
#19 randomly [tiab]
#20 trial [tiab]
#21 groups [tiab]
#22 #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19
#13 and #22

Em base (OVID) search strategy

#1 Exp Tinnitus/

#2 tinnit*.tw.

#3 (EAR* and (BUZZ* or RING* or ROAR* or CLICK* or PULS*)).ti.

#4 1 or 3 or 2

#5 (CBT or ACT or mindfulness or MBTR or MBSR or MBTSR or psychoeducation or iACT or iCBT or GCBT)

#6 ((cogniti* or relaxation or acceptance or commitment or adaptation) adj2 (therap* or behavior* or behaviour* or strateg* or intervention* or approach* or psychotherap* or training or treatment or technique* or program* or counseling or counselling))

#7 ((behaviour* or behavior* or meditation) adj2 (strateg* or intervention* or therap* or approach* or psychotherap* or technique* or counseling))

#8 desensiti* and psychology*

#9 interpersonal therapy

#10 6 or 7 or 5 or 8 or 9

#11 10 and 4

#12 randomized controlled trial [pt]

#13 controlled clinical trial [pt]

#14 randomized [tiab]

#15 placebo [tiab]

#16 drug therapy [sh]

#17 randomly [tiab]

#18 trial [tiab]

#19 groups [tiab]

#20 #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19

#21 #11 and #20

Medline (OVID) Search Strategy

#1 tinnitus.mp. or exp TINNITUS/

#2 randomi?ed control* trial.mp.

#3 cognitive therapy.mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]

#4 behavio?ral therapy.mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]

#5 (cognit* or desensiti* or behav*).mp.

#6 "Conditioning (Psychology)"/

#7 RELAXATION/

#8 (therap* or train* or modification).mp.

#9 Psychotherapy/ or psychotherap*.mp. or Cognitive Therapy/

#10 treatment.mp. or Therapeutics/

#11 (Ear* and (buzz* or ring* or Roar* or Click* or PULS*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]

#12 (crossover procedure or double-blind procedure or single-blind procedure or (random* or factorial* or crossover* or placebo* or doubl* blind* or singl* blind*)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]

#13 mindfulness.mp. or exp MINDFULNESS/ or exp Cognitive Therapy/ or exp Relaxation Therapy/

#14 exp "ACCEPTANCE AND COMMITMENT THERAPY"/ or acceptance.mp.

#15 exp Psychotherapy/ or interpersonal therapy.mp. or exp Psychotherapy, Brief/ or exp Cognitive Therapy/

#16 (MBTR or MBSR or MBTSR or psychoeducation or iACT or iCBT or GCBT).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
#17 ((cogniti* or relaxation or acceptance or commitment or adaptation) adj2 (therap* or behavior* or behaviour* or strateg* or intervention* or approach* or psychotherap* or training or treatment or technique* or program* or counseling or counselling)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
#18 ((behaviour* or behavior* or meditation) adj2 (strateg* or intervention* or therap* or approach* or psychotherap* or technique* or counseling or counselling)).mp. [mp=title, abstract, original title, name of substance word, subject heading word, keyword heading word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]
#19 1 or 11
#20 2 or 12
#21 5 or 6 or 7
#22 8 or 9 or 10
#23 21 and 22
#24 3 or 4 or 13 or 14 or 15 or 16 or 17 or 18 or 23
#25 19 and 20 and 24

PsychInfo (OVID) Search Strategy

#1 tinnitus.mp. or exp TINNITUS/
#2 randomi?ed control* trial.mp.
#3 (EAR* and (BUZZ* or RING* or ROAR* or CLICK* or PULS*)).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#4 exp Group Psychotherapy/ or exp Cognitive Therapy/ or exp Cognitive Behavior Therapy/ or exp Behavior Therapy/ or behavio?ral therapy.mp.

#5 ((cognit* and behav*) or (DESENSITI* and PSYCHOLOG*) or (IMPLOSIVE* and THERAP*)).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#6 behavi?r modification/ or behavi?r therapy/ or cognitive therapy/

#7 ((COGNIT* or BEHAV* or CONDITIONING or RELAXATION or DESENSITI*) and (THERAPY or THERAPIES or THERAPEUTIC* or PSYCHOTHERAP* or TRAIN* or RETRAIN* or TREATMENT* or MODIFICATION*)).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#8 cognitive behavio?ral therapy.mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#9 tinnitus.mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#10 1 or 3 or 9

#11 4 or 7 or 8

#125 or 6 or 11

#13 (crossover procedure or double-blind procedure or single-blind procedure or (random* or factorial* or crossover* or placebo* or doubl* blind* or singl* blind*)).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#14 2 or 13

#15 exp "ACCEPTANCE AND COMMITMENT THERAPY"/ or acceptance.mp.

#16 mindfulness.mp. or exp MINDFULNESS/

#17 exp psychotherapy/ or exp "acceptance and commitment therapy"/ or exp behavior therapy/ or exp cognitive therapy/

#18 exp stress management/ or exp anxiety management/ or exp behavior modification/ or exp cognitive techniques/ or exp stress/ or exp treatment/

#19 15 or 16 or 17 or 18

#20 12 or 19

#21 interpersonal therapy.mp. or exp Interpersonal Psychotherapy/

#22 ((behaviour* or behavior* or meditation) adj2 (strateg* or intervention* or therap* or approach* or psychotherap* or technique* or counseling or counselling)).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#23 ((cogniti* or relaxation or acceptance or commitment or adaptation) adj2 (therap* or behavior* or behaviour* or strateg* or intervention* or approach* or psychotherap* or training or treatment or technique* or program* or counseling or counselling)).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#24 (CBT or ACT or mindfulness or MBTR or MBSR or MBTSR or psychoeducation or iACT or iCBT or GCBT).mp. [mp=title, abstract, heading word, table of contents, key concepts, original title, tests & measures]

#25 20 or 21 or 22 or 23 or 24

#26 10 and 14 and 25

PubMed Search Strategy

#1 "Tinnitus" [Mesh]

#2 tinnit* [tiab]

#3 (EAR* [ti] AND (BUZZ* [tiab] OR RING* [tiab] OR ROAR* [tiab] OR CLICK* [tiab] OR PULS* [tiab]))

#4 #1 OR #2 OR #3

#5 CBT* [tiab] OR ACT* [tiab] OR MINDFULNESS* [tiab] OR MBTR* [tiab] OR MBTSR* [tiab] OR PSYCHOEDUCATION* [tiab] OR IACT* [tiab] OR ICBT* [tiab] OR GCBT* [tiab]

#6 (COGNITI* [tiab] OR RELAXATION* [tiab] OR ACCEPTANCE* [tiab] OR COMMITMENT* [tiab] OR ADAPTATION* [tiab]) near (THERAP* [tiab] OR BEHAVIOR* [tiab] OR BEHAVIOUR* [tiab] OR STRATEG* [tiab] OR INTERVENTION* [tiab] OR APPROACH* [tiab] OR PSYCHOTHERAP* [tiab] OR TRAINING* [tiab] OR TREATMENT* [tiab] OR TECHNIQUE* [tiab] OR PROGRAM* [tiab] OR COUNSELING* [tiab] OR COUNSELLING* [tiab])

#7 (BEHAVIOUR* [tiab] OR BEHAVIOR* [tiab] OR MEDITATION* [tiab]) near (STRATEG* [tiab] OR INTERVENTION* [tiab] OR THERAP* [tiab] OR APPROACH* [tiab] OR PSYCHOTHERAP* [tiab] OR TECHNIQUE* [tiab] OR COUNSELLING* [tiab]

#8 DESENSITI* [tiab] AND PSYCHOLOG* [tiab]

#9 INTERPERSONAL THERAPY* [tiab]

#10 #5 OR #6 OR #7 OR #8 OR #9

#11 Randomized controlled trial [td]

#12 Controlled clinical trial [pt]

#13 Randomized [tiab]

#14 Placebo [tiab]

#15 Drug therapy [sh]

#16 Randomly [tiab]

#17 Trial [tiab]

#18 Groups [tiab]

#19 #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18

#20 #4 and #10 and #19

Appendix 7. Data Items

Data items included information on: the source of the study, the eligibility criteria (including reasons for exclusion), the participants (total number, diagnostic criteria, mean duration of tinnitus, % of bilateral tinnitus, age, sex, country and comorbidities), the study methods (study design, duration and risk of bias elements), the interventions (number of arms in the trial, classification of each arm, protocol used, number of sessions, duration of treatment, delivery method, follow-up duration), the outcomes (for each outcome of interest: validated questionnaire used), the results (for each outcome of interest: number of participants allocated to each intervention, missing participants, means and SDs for pre/post/ follow-up results, estimate of effects with confidence intervals) and the key points from the conclusions.

A copy of the data extraction form can be found in the accompanying excel document.

Appendix 8. Risk of Bias in Individual Studies

Risk of bias assessment, as guided by the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2011), will be performed on all included studies.

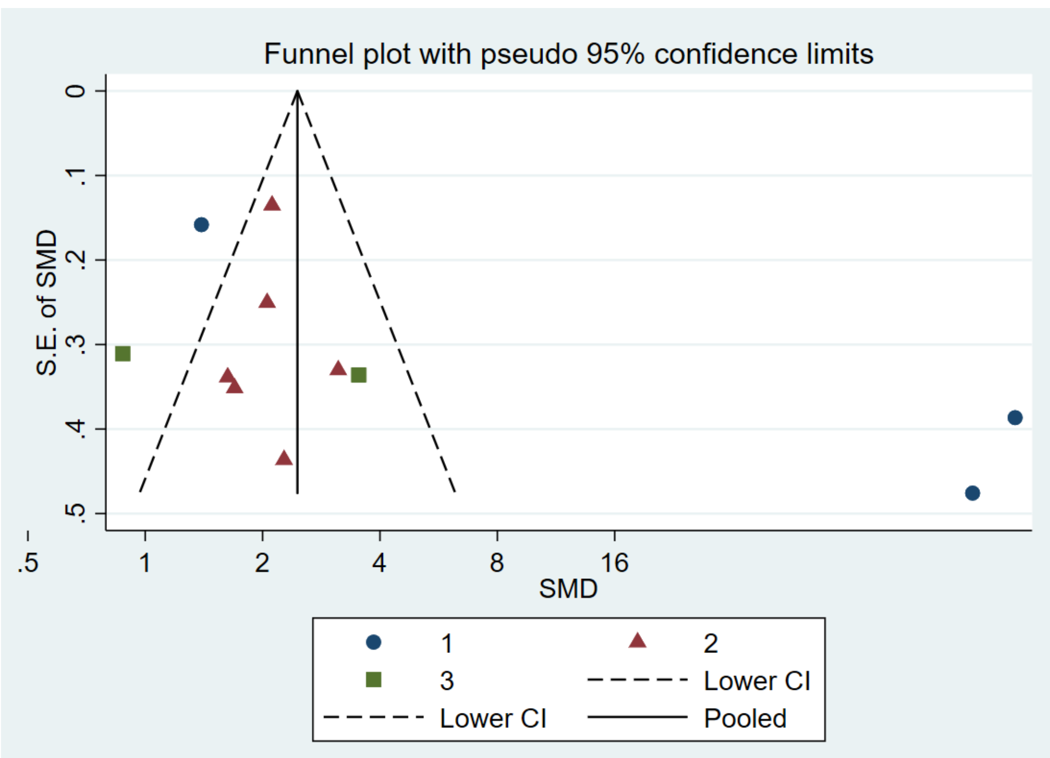
The following risks of bias will be assessed:

1. sequence generation;
2. allocation concealment;
3. blinding;
4. incomplete outcome data;
5. selective outcome reporting; and
6. other sources of bias.

Appendix 9. Funnel Plot

Funnel plot of the subgroup meta-analysis for tinnitus HRQOL. The asymmetry observed may potentially be due to publication bias.

Figure 13 . Funnel Plot for tinnitus HRQOL meta-analysis



1- Self-administered CBT; 2- Group CBT; 3- Cognitive/Behavioral ‘only’ therapies

Appendix 10. Funding

Sources of Funding

No financial support was obtained for the completion of this study.

Declaration of Conflicts of Interest

The author(s) declared no conflicts of interest.

Registration

Registration with PROSPERO will be undertaken.

Appendix 11. PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	3
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5-7
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	8
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	41
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	9-10

Section/topic	#	Checklist item	Report on pag #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9-10
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	10
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	11-14
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	15
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	17-27
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	17-27
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	48
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	17-27
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	28
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	29
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	30
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	49

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097 For more information, visit: **www.prisma-statement.org**